AI for Manufacturing GRCC Training

AI for Workforce

intel. digital readiness

Welcome!

Get all the content at:
https://github.com/grcc-resendiz/ai-manufacturing-muskegon

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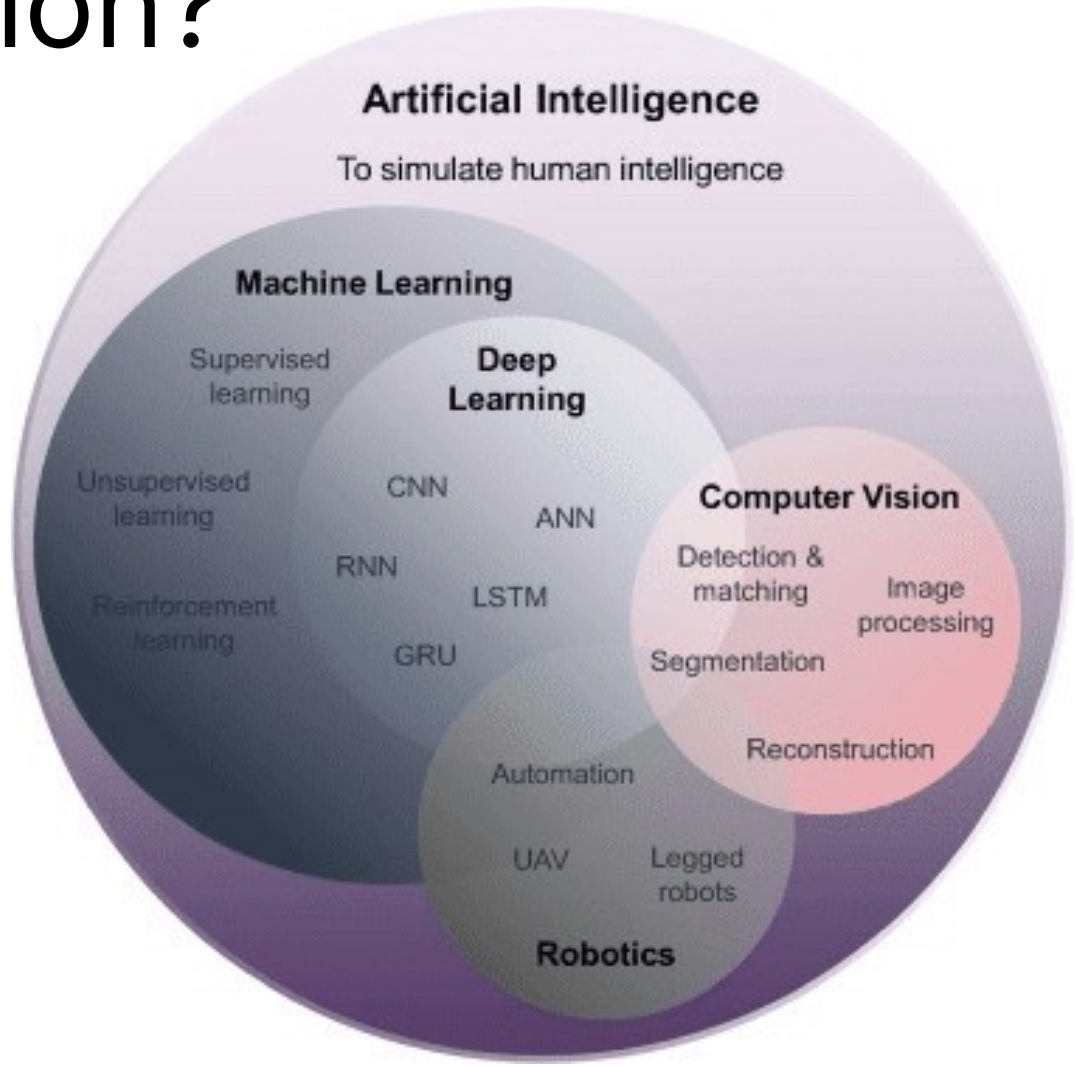
Day 2 Agenda

- **0:00** Welcome and Day 1 recap
- **0:10** Computer vision in manufacturing
- **0:40** Hands on: train a model to spot defective cast parts
- **1:25** Debrief and your first pilot
- **1:40** Break
- **1:45** Vibe coding: build a simple app by describing it
- **2:50** Next session and questions

What is Computer Vision?

What is Computer Vision?

Computer vision is a subset of Artificial Intelligence that trains computers to automatically process, extract, and manipulate visual data from images and videos.



What is Computer Vision?

Common tasks that CV can perform:

- Object identification
- Object classification
- Image segmentation

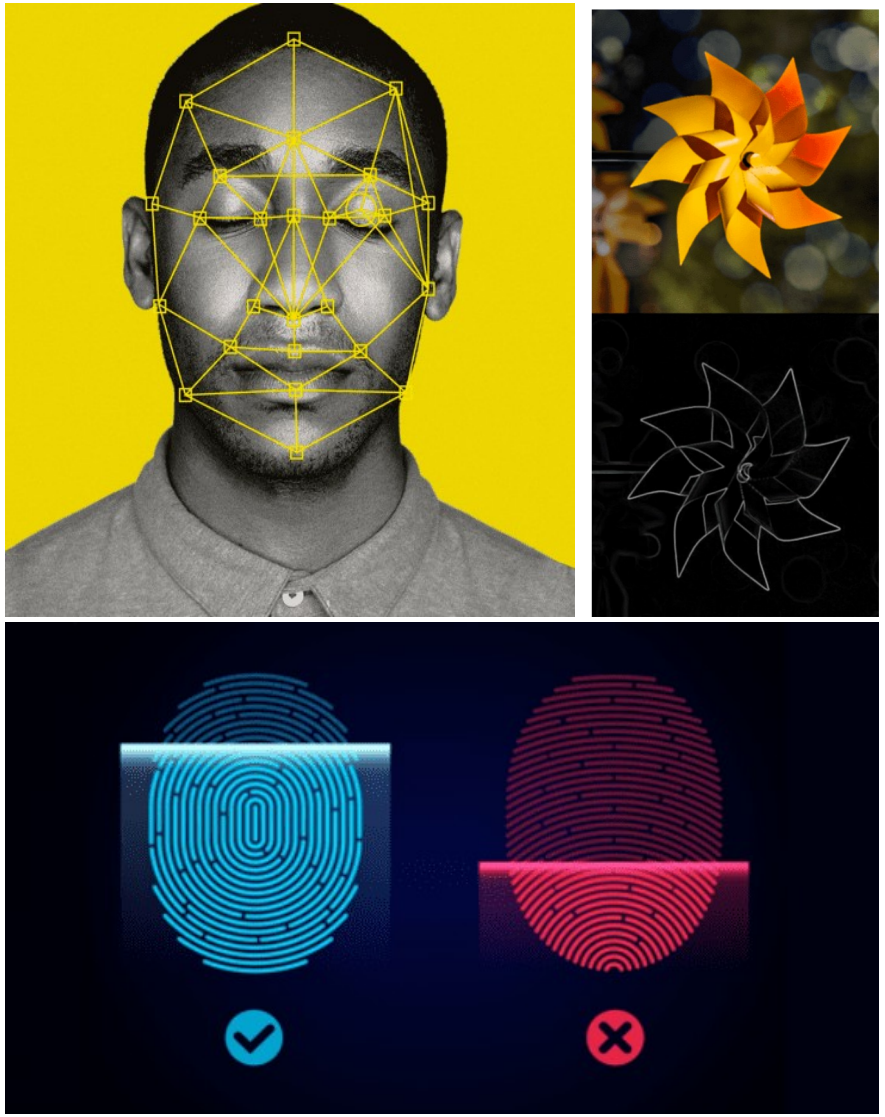




What is Computer Vision?

Common tasks that CV can perform:

- Facial recognition
- Edge Detection
- Pattern recognition



Computer Vision in the Manufacturing Industry

Existing Use Cases - Quality Inspection

- One of the most common use cases of computer vision in manufacturing is to inspect and validate the quality of products during the manufacturing process.

- This involves using computer vision algorithms to detect defects and deviations from specifications.

- **Why defect detection is hard?**



Image source : piotr.sember@uico.pl. (2022, August 24). *Quality inspections of the production line using computer vision and Flow* –. Novacura.

https://www.novacura.com/computer-vision-quality-inspections/

Computer Vision in the Manufacturing Industry

Existing Use Cases - Quality Inspection

- **What makes CV valuable**
- More consistent inspection
- Faster decision-making
- Scalable across shifts and sites
- Better traceability
- Supports operators rather than replacing judgment



Image source : piotr.sember@uico.pl. (2022, August 24). *Quality inspections of the production line using computer vision and Flow* –. Novacura.

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Computer Vision in the Manufacturing Industry

Existing Use cases - Object Recognition and Sorting

- Computer vision is used to recognize objects and sort them into categories, such as by size, shape, color, or material.
- This can help to streamline the manufacturing process and improve efficiency.



Atos Group. (2017, June 19). *Predictive Maintenance for Manufacturing*. YouTube. https://www.youtube.com/watch?v=QStfd20NH3I

Computer Vision in the Manufacturing Industry

Existing Use cases - Robotic Guidance and Automation

- Computer vision is used to guide robots and automate tasks in the manufacturing process, such as pick-and-place operations or assembly tasks.

- This helps to improve efficiency and reduce the need for manual labor.



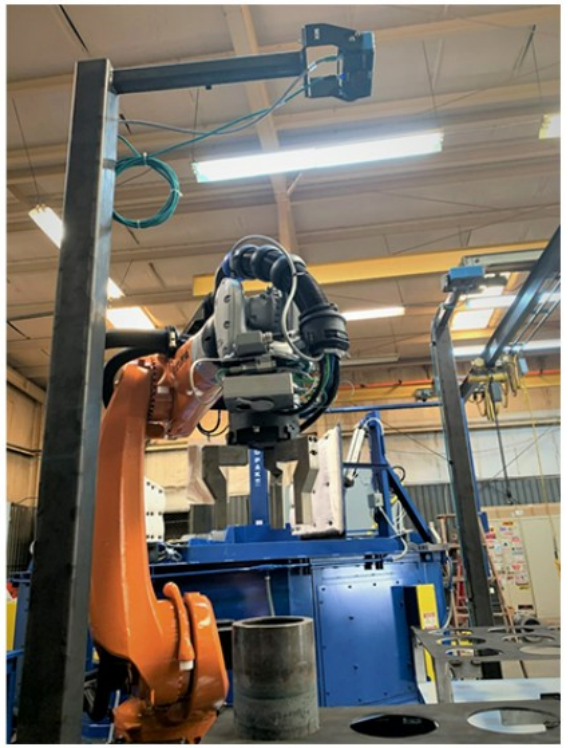
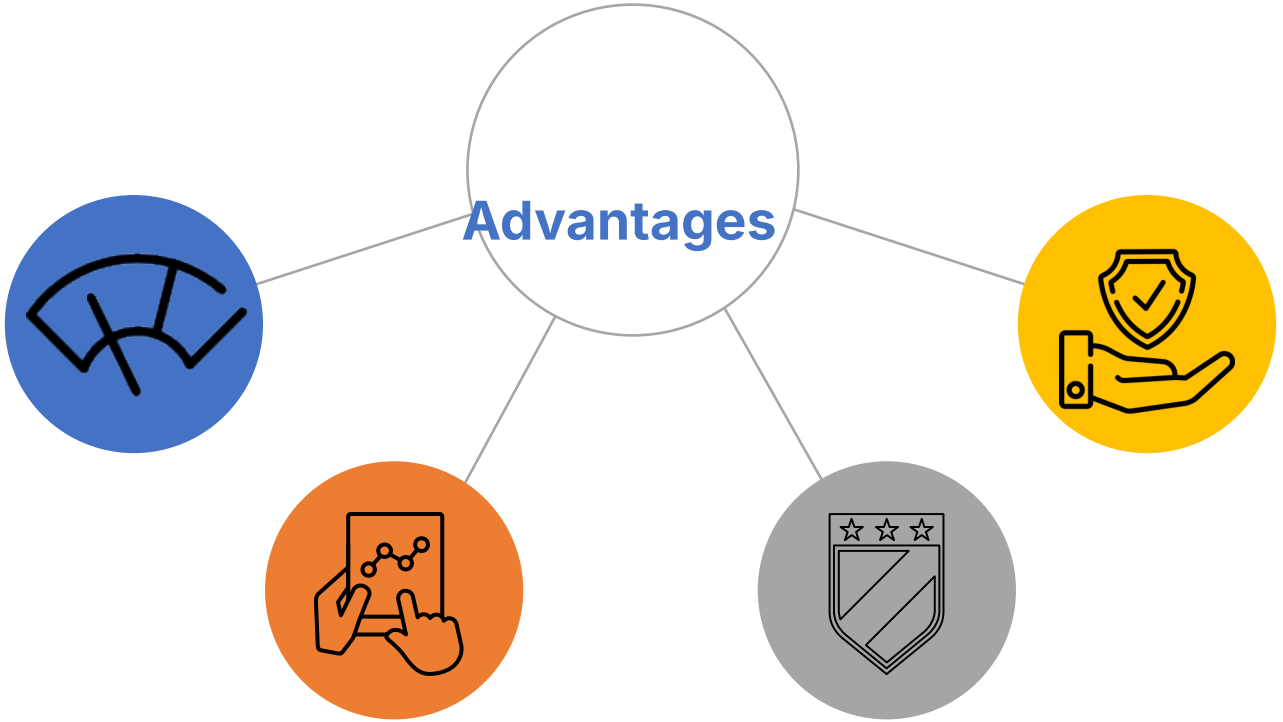


Image source : SICK USA. (2020, December 19). *Robot Vision and Guidance Can Take Automation to the Next Level*. SICK USA BLOG. https://sickusablog.com/robot-vision-guidance-can-take-automation-next-lev

Computer Vision in the Manufacturing Industry

Advantages



Increased Efficiency
By automating repetitive tasks, computer vision can
increase manufacturing speed and efficiency while reducing labor costs.

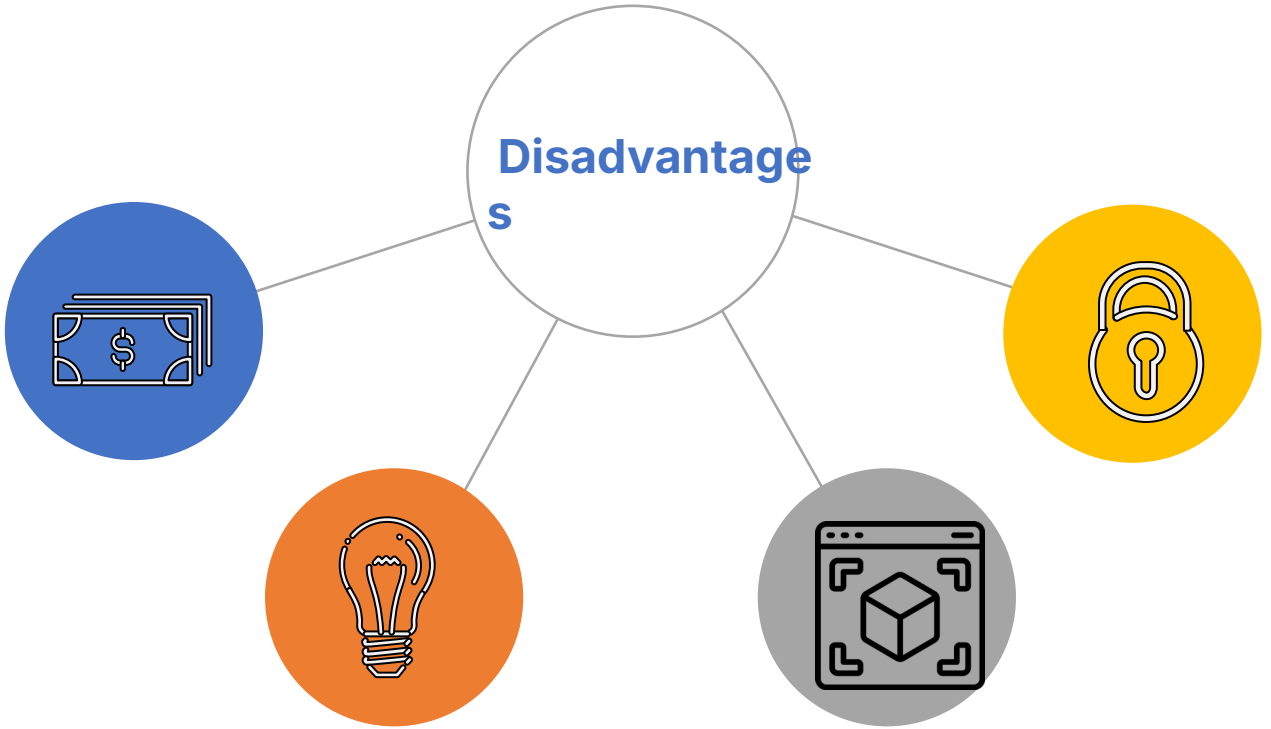
Reliability
The computer system's effectiveness remains constant despite limitations like fatigue or disease.

Better Data Collection and Analysis
Computer vision technology can collect and analyze large amounts of data in real-time, providing manufacturers with important insights into their processes and operations.

Safety
Computer vision technology can help detect potential safety hazards in the manufacturing process, making a secure work environment for employees.

Computer Vision in the Manufacturing Industry

Disadvantages



High Initial Cost
Implementing computer vision technology in manufacturing requires significant investment in hardware, software, and personnel training.

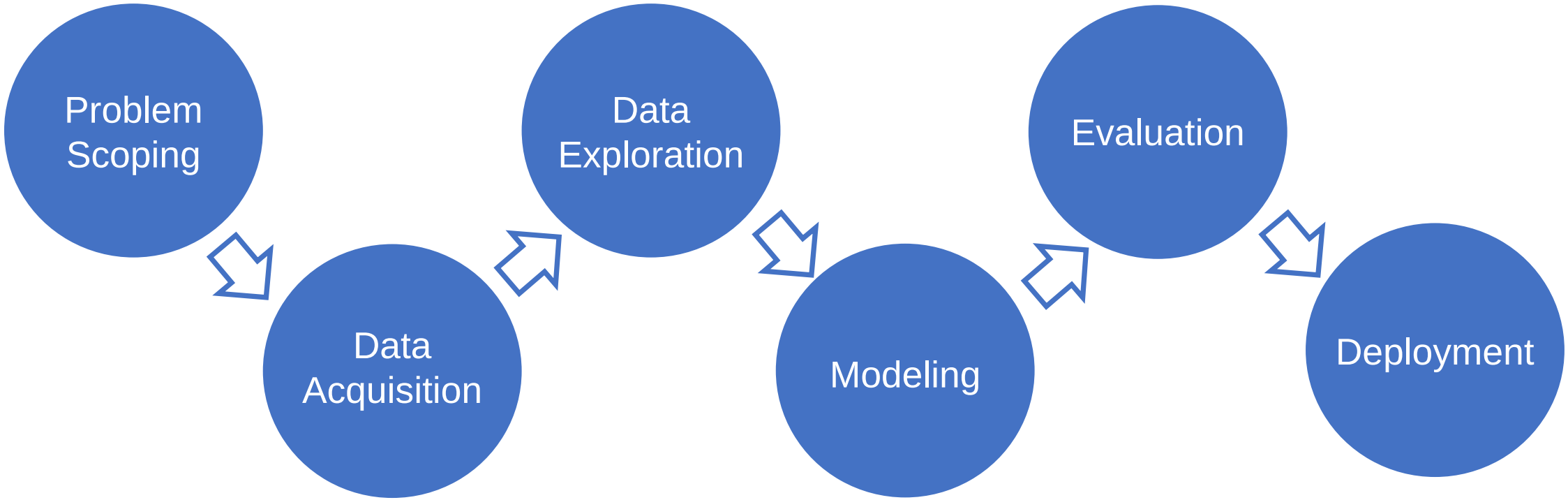
Privacy Concerns
Some employees may have concerns about their privacy when being monitored by computer vision systems, leading to potential pushback or resistance to adoption.

Dependence on Lighting and Environment
Computer vision systems require consistent lighting conditions and a clear view of the manufacturing process to work accurately.

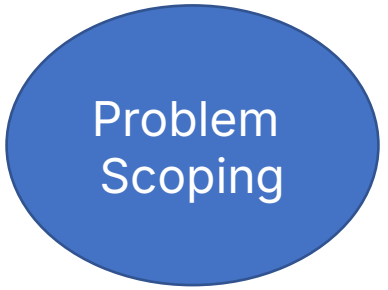
Limitations in Object Recognition
Computer vision may have limitations in recognizing complex objects or identifying subtle differences in product features, leading to inaccuracies in detecting defects.

Hands on Activity

Let’s Recall the AI Project Cycle



Hands - On Activity



Teachable Machine Image Project Development

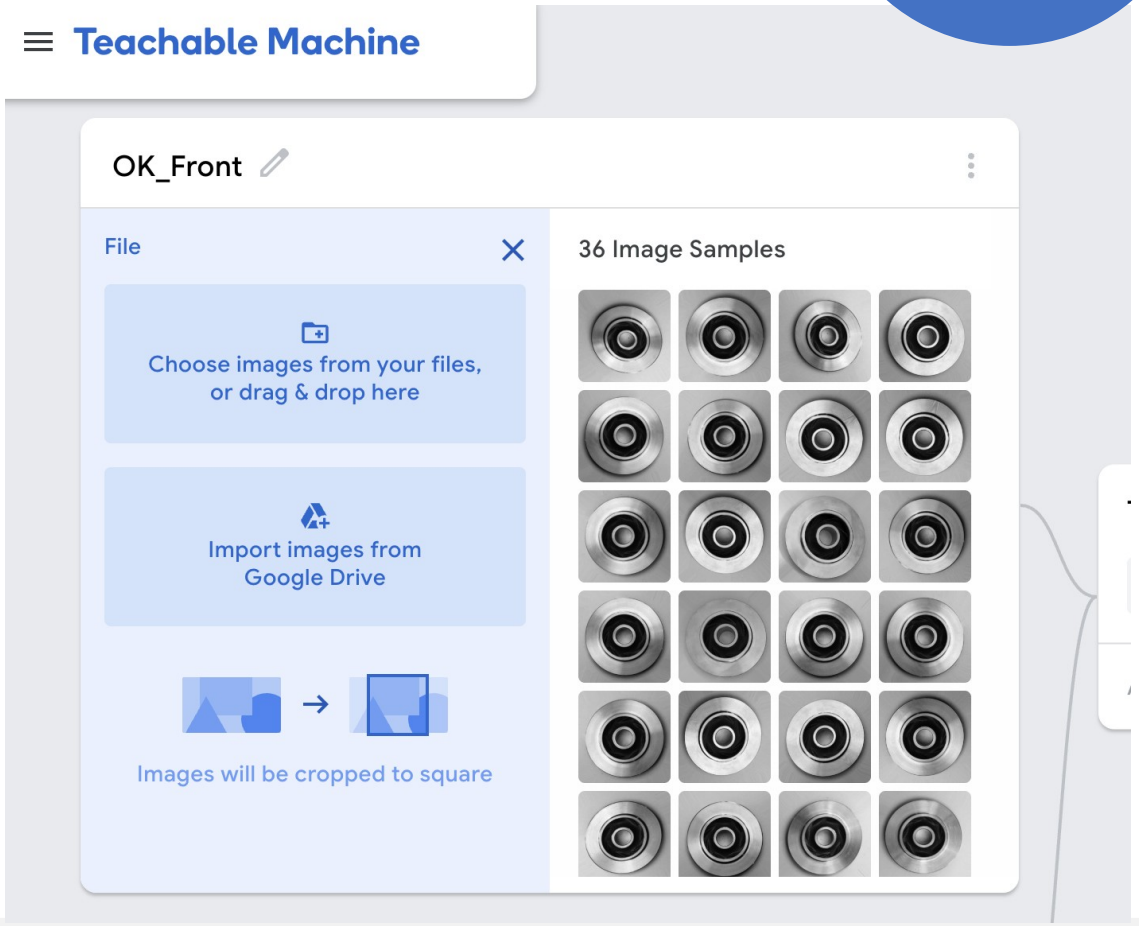
- Train a model that tells OK cast parts from defective ones.
- Download casting_data.zip from the course GitHub page and unzip it.

Hands - On Activity

Teachable Machine Image Project Development

Data Acquisition

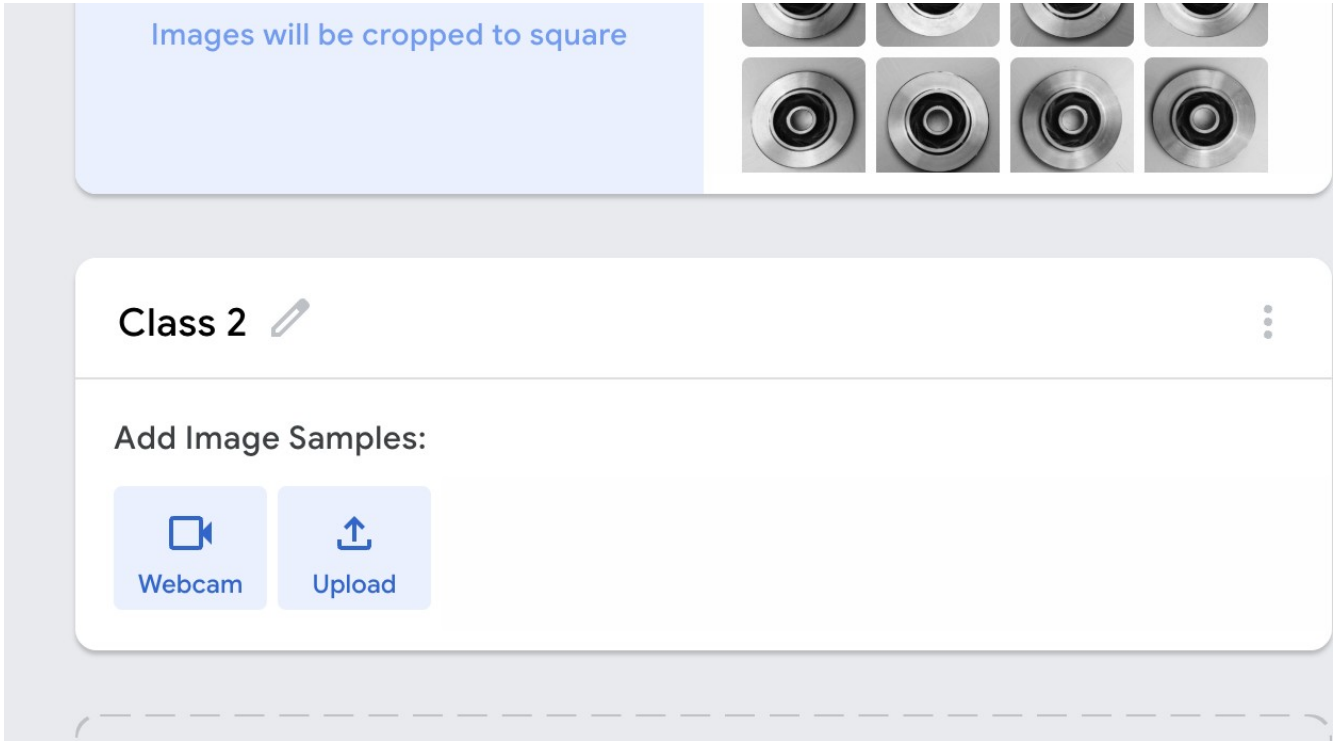
- Click the pencil icon and rename Class 1 to OK.
- Upload the images from train/ok_front.



Hands - On Activity

Teachable Machine Image Project Development

- Rename Class 2 to Defective.
- Click Upload and add the images from train/def_front.

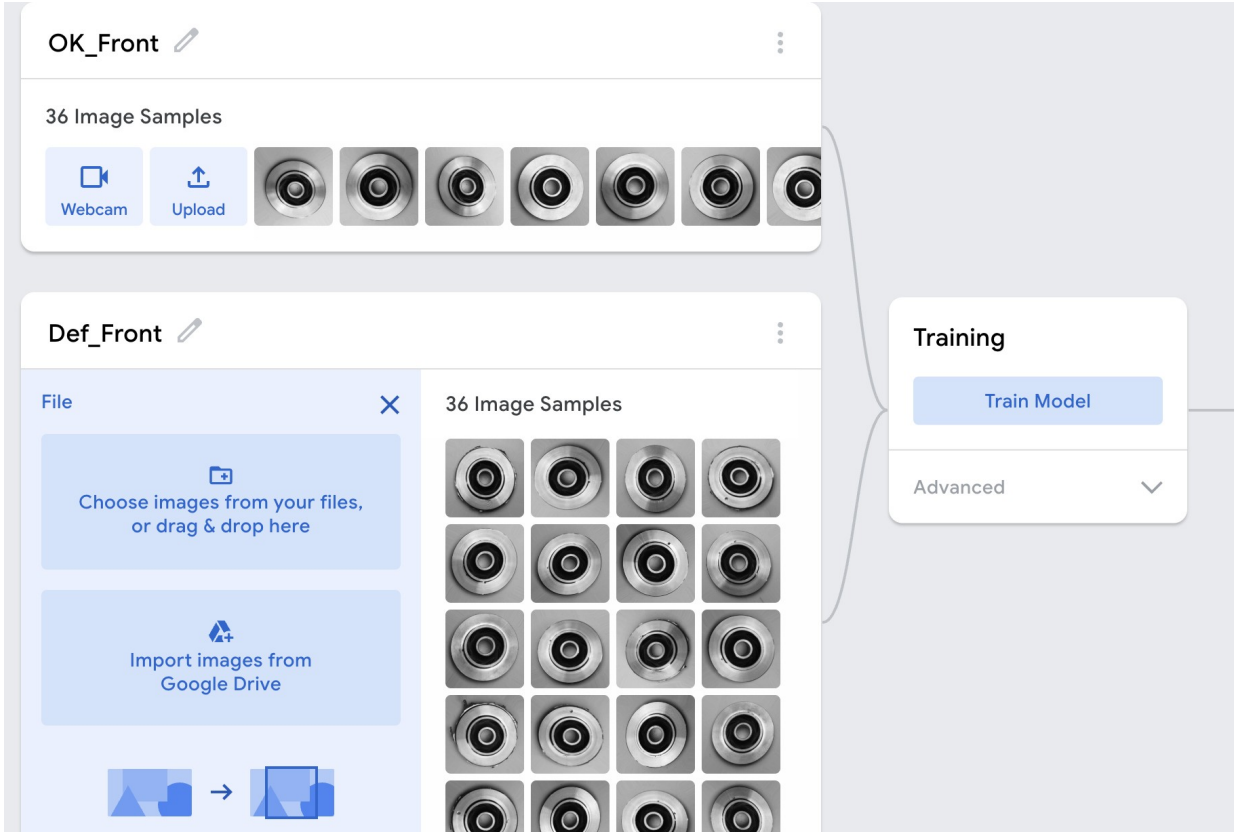


Hands - On Activity

Modeling

Teachable Machine Image Project Development

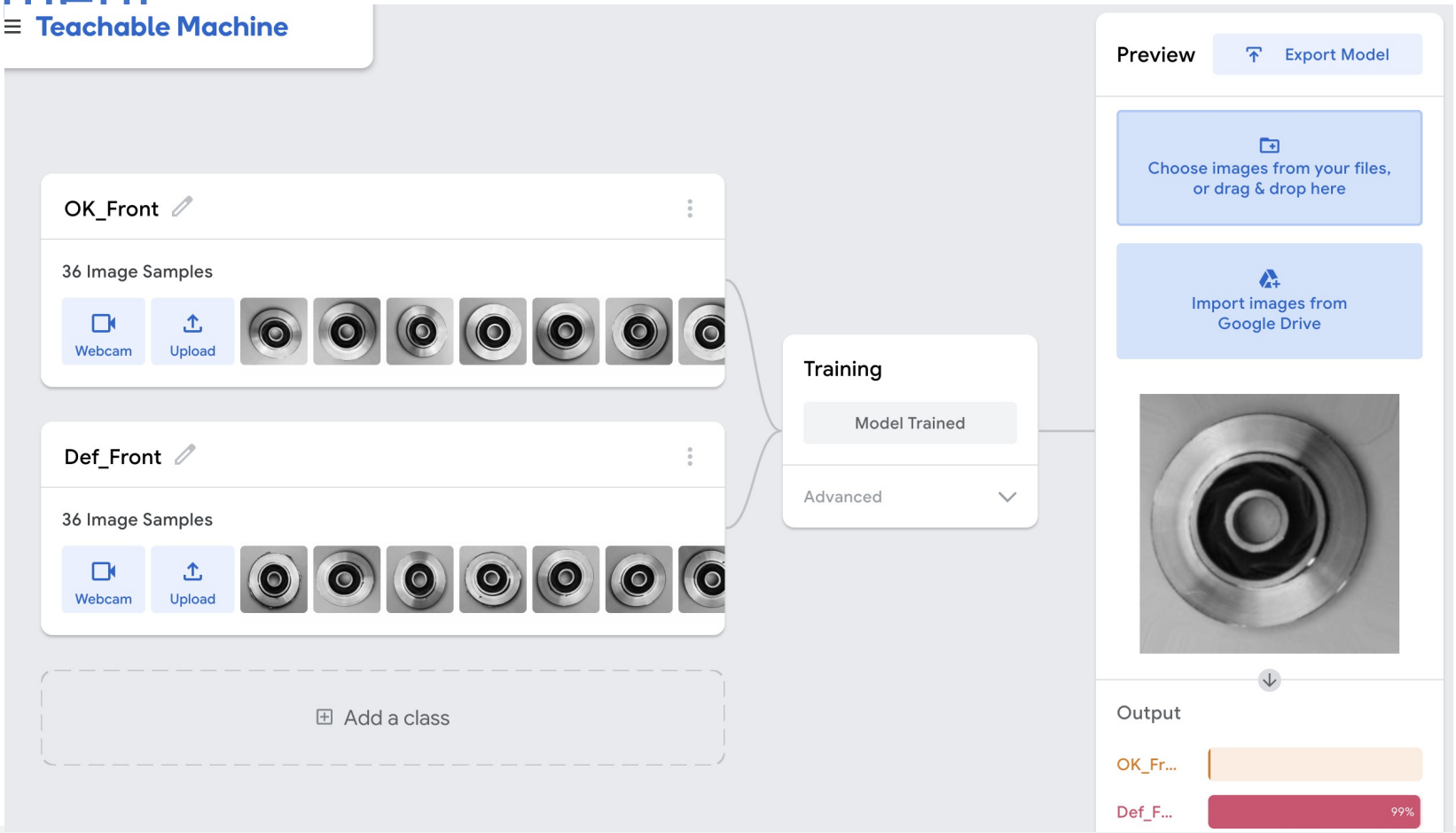
- Click Train Model.
- Keep the browser tab open while it trains.



Hands - On Activity

Teachable Machine Image Project Development

Evaluate

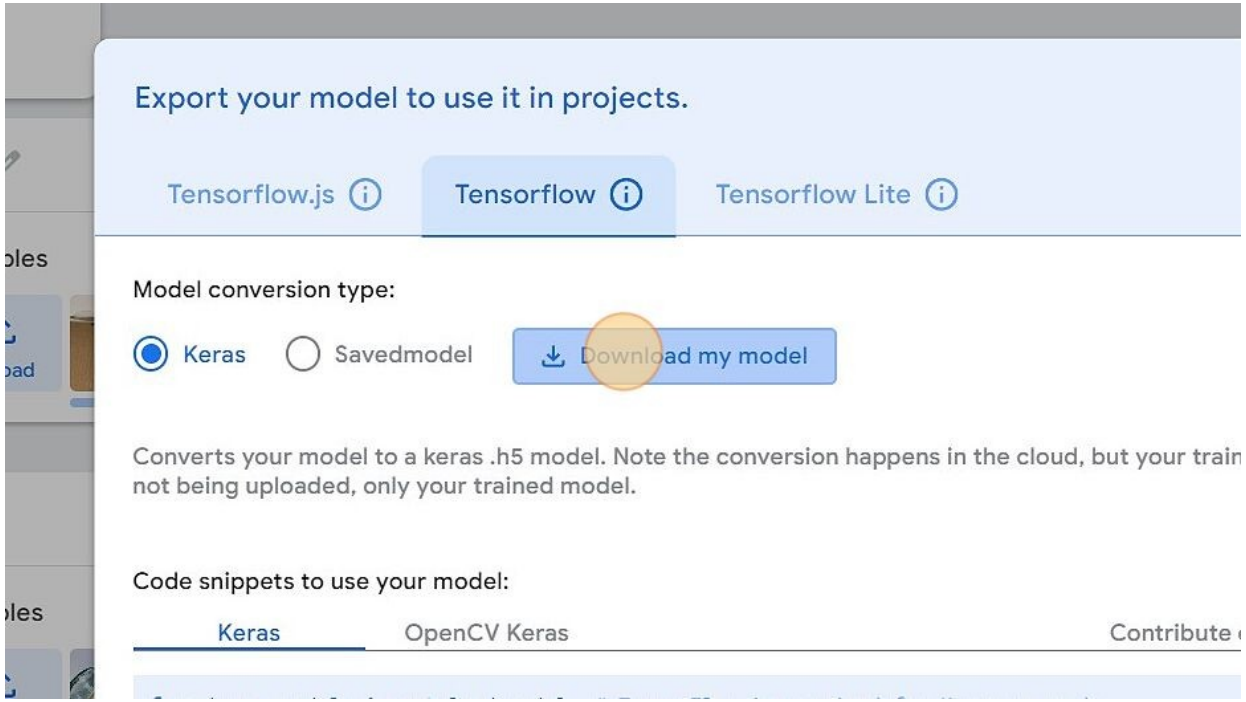


Hands - On Activity

Teachable Machine Image Project Development

Deployment

- Click Export Model, then Upload my model.
- Copy the shareable link.
- Paste it into the Codespaces demo to test with your webcam.



Reflection

Hands - On Activity

Teachable Machine Image Project Development

- **Debrief**
- What did your model get right—and wrong?
- Which defects were easiest?
- Which cases confused the model?
- Did lighting matter?
- Did data balance matter?

Technology Adoption Gap

- Questions for a first pilot
- Which defect matters most?
- What images do we already have?
- What decision will the system trigger?
- What error is more costly: miss or false alarm?
- Who owns data, labeling, and validation?
- How will we measure pilot success?

Key Takeaways

- Start small, learn fast, scale what works
- Don’t begin with enterprise-wide transformation
- Begin with one painful inspection point
- Use data discipline
- Build confidence with a pilot
- Let operations lead the use case

Vibe Coding

What Is Vibe Coding?

- You describe the app in plain language.
- The AI writes the code and shows you a working preview.
- You test it and ask for changes until it does the job.
- Great for quick prototypes. Not a replacement for IT or production software.

Prompt, Skill, or App?

- **Prompt:** a one time question or task
- **Skill:** the same question again and again, using your documents
- **App:** a repeated workflow with data in and a result out, like a dashboard
- Start with the simplest one that solves the problem.

Demo: Defect and Downtime Dashboard

- **Data:** manufacturing_data_export_2026-05-01.csv (course GitHub page)
- **Columns:** date, machine, quantity, defects, downtime, energy, operator, shift, revenue, cost
- **Goal:** see which machine and shift lose the most time and parts
- **Tool:** Google AI Studio, Build mode (aistudio.google.com/apps)

The Prompt

- Build a web app for a manufacturing supervisor.
- Let me upload a CSV with these columns: date, machineId, productionQuantity, defects, downtimeMinutes, energyConsumption, operator, shift, revenue, cost.
- Show defect rate and downtime by machine and by shift as bar charts.
- List the top 3 problems in plain language with one suggested action each.
- Keep it simple, one page, large text.

Build It in Google AI Studio

- Go to aistudio.google.com/apps and sign in with a Google account.
- Paste the prompt and run it. The code and a live preview appear.
- Upload the CSV in the preview and check the numbers.
- Ask for one change in the chat, for example: add a filter by date.
- Claude, ChatGPT, and Gemini can also build small apps right in the chat.

Map Your Workflow

- Is the task part of a bigger workflow?
- Can you write the steps down clearly?
- Are the inputs and outputs the same every time?
- Is it simple enough to build without IT help?
- Rate your tasks on time spent, frequency, and repetition. High on all three is your best app idea.

Build Your Own

- Pick one task from your worksheet.
- Write the prompt with RTFC: who it is for, what it does, what it shows, and the limits.
- Use made up or sample data. Never upload confidential company data.
- Start small, test with real questions, and ask for one change at a time.

Prototype Is Not Production

- Check the results against the real numbers.
- Ask who will maintain it and where the data lives.
- Bring IT in before real company data goes in.
- A working prototype is a great way to start that conversation.

Next Session: Predictive Maintenance

- You will predict machine failures from sensor data.
- You will build the model in Orange, no coding needed.
- You will plan your first AI pilot.

Questions?


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